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using System;
using System.Collections.Generic;
using System.Drawing;
using System.Text;

namespace Lab3_KLeach
{
    // This program is a C# console application that plays a hard coded game of Hangman with the user.
    // The word the player will be trying to guess is "Sequence".
    // This program will do the following:
    // Print a basic interface showing the player the "blanks"
    // Display how many extra chances the player has left before the game is over.
    /* A message will display if the player has guessed the word correctly or if
    // they have run out of guesses.*/
    // A CurrentWord class handles the logic for checking if a guess is correct
    // and returning the current state of the word being guessed.
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    /* Instead of a hard coded word, the program has an array of 15 words, all with length less than 10.
    Generate a random number to select one of the words.*/

    internal class Program
    {
        static void Main(string[] args)
        {
            // Array of 15 words to guess from
            string[] words = { "Sequence", "Benevolent", "Jurassic", "Diagram", "Haboob", "Traffic", "Harness", "Fraction", "Lasso",
"Wilderness", "Kilogram", "Global", "Quality", "Spatial", "Pneumonic" };
            Random random = new Random();
            string selectedWord = words[random.Next(words.Length)]; //Selects a random word from the array
            // Create a new CurrentWord object using the player's input.
            // The CurrentWord class will handle the logic for checking if a guess is correct
            CurrentWord currentWord = new CurrentWord(selectedWord);
            int remainingGuesses = 6; // Number of allowed incorrect guesses
            // Create a HashSet to store every letter the player has already guessed.
            // A HashSet is used because it automatically prevents duplicate entries
            // and makes checking "has this letter been guessed already?" very fast
            HashSet<char> guessedLetters = new HashSet<char>();
            // Print a welcome message to greet the player
            Console.WriteLine("Welcome to Hangman!");
            while (remainingGuesses > 0 && !currentWord.IsComplete())
            {
                Console.WriteLine($"Current word: {currentWord.GetDisplayedWord()}");
                Console.WriteLine($"Remaining guesses: {remainingGuesses}");
                // Prompt the player to enter the word/letter they will be guessing
                Console.Write("Enter your guess (single letter): ");
                string? input = Console.ReadLine();
                if (string.IsNullOrEmpty(input))
                {
                    Console.WriteLine("No input detected, try again.");
                    continue;
                }
                char guess = input[0];
                if (currentWord.CheckGuess(guess))
                {
                    Console.WriteLine("You got it!");
                }
                else
                {
                    remainingGuesses--;
                    Console.WriteLine("Nope, try again.");
                }
            }
            if (currentWord.IsComplete())
            {
                Console.WriteLine($"You Win!");
            }
            else
            {
                Console.WriteLine($"You Lose!");
            }
            Console.WriteLine("Do you feel lucky, punk? (y/n)");
            string? playAgain = Console.ReadLine();
            if (!string.IsNullOrEmpty(playAgain) && playAgain.Trim().Equals("y", StringComparison.OrdinalIgnoreCase))
            {
                Main(args); // Restart the game
            }
        }
    }
}

```